

Root cause analysis RCA_2026_014: high speed stage failure of GBX_A17_02

Northwind Energy | asset WTG_A17 | prepared by Priya Raman | 10 April 2026 | classification: internal | synthetic sample document

Scope and summary

This root cause analysis covers the high speed stage failure of gearbox GBX_A17_02, installed in turbine WTG_A17 at the Esbjerg Nord site operated by Northwind Energy in the Danish North Sea. The turbine sits on the SUB_NORD_02 collector circuit. On 14 March 2026 the condition monitoring system recorded a broadband velocity of 5.9 mm/s RMS at the high speed stage, above the 4.5 mm/s alarm limit that standard NW_STD_0417 defines, and the unit was removed from service the same day. Borescope inspection and subsequent strip down found advanced spalling on the upwind planet bearing outer raceway. The gearbox was supplied by Halvard Drivetrain and the strip down was witnessed by their field engineers at Esbjerg, with hardness coupons tested by an independent laboratory. This report establishes the failure mechanism, explains why the alarm fired when it did rather than earlier, and identifies the contributing factor that brought the failure forward of design life. It was prepared and signed by Priya Raman, lead reliability engineer, Northwind Energy. The findings apply to one unit, but the contributing factor does not, which is the reason this report ends with a fleet action rather than a single repair order.

Sequence of events

The unit ran without exception through the 2025 autumn campaign. Work order WO_88231 recorded a routine oil change and filter inspection on 2025-11-04, carried out by Anders Bjerg; the filter debris was noted as light and within limits at that time, which in hindsight was the first observable symptom rather than a clean result. Through January 2026 the trended velocity at the high speed stage rose slowly from 2.1 mm/s to 2.8 mm/s. That rise stayed below the 3.2 mm/s warning level, so it raised no automatic flag, and because the increase was gradual it did not trigger a rate of change notification either. In the last week of February the spectrum changed character: the overall level barely moved, but discrete sidebands appeared around the gear mesh frequency. On 11 March the overall level began to climb steeply, reaching 5.9 mm/s on 14 March, at which point the alarm fired and the turbine was stopped. The elapsed time from the first clear spectral signature to the alarm was about two weeks, and the elapsed time from the alarm to removal was under twelve hours.

Why the bearing failed

The mechanism established for GBX_A17_02 on strip down is subsurface initiated rolling contact fatigue, progressing to macroscopic spalling. It runs in a sequence, and each step explains the next. Under load the rolling elements produce a Hertzian contact stress field whose maximum shear stress sits a short distance below the raceway surface, not at it. Repeated passes drive microstructural change at that depth, and micro cracks initiate there. Those cracks propagate toward the surface, and when they break through, a flake of material detaches and leaves a pit. That first pit is the important event, because the failure is self accelerating from that moment on. The pit destroys the local contact geometry, so the rolling element now strikes an edge instead of rolling over a continuous curve. The impact raises the local contact stress far above nominal, which initiates further cracks at the pit boundary, which enlarges the pit. Detached debris also circulates in the oil and work hardens or dents other surfaces, seeding new initiation sites elsewhere. In GBX_A17_02 the debris had already dented the downwind planet raceway, which is why the whole unit was replaced rather than the failed bearing alone when it came out of WTG_A17. The process is not linear in time: it is slow and nearly invisible until the first through crack, then progressively faster. Any monitoring scheme that only watches an overall level is watching the wrong part of that curve, because the overall level does not move appreciably until the damage is already extensive, and that is precisely the trap NW_STD_0417 is written to avoid.

Why the alarm fired when it did

Spalling has a distinctive spectral fingerprint, and understanding it explains both the two week warning and the previous months of silence. Each time a rolling element passes over the damaged zone it produces a short impulse. Those impulses repeat at the bearing pass frequency, which is set by geometry and shaft speed, and they amplitude modulate the much stronger gear mesh vibration. In the frequency domain, amplitude modulation does not appear as a peak at the modulating frequency. It appears as pairs of sidebands straddling the carrier, offset by the modulating frequency, which here means pairs around the gear mesh frequency spaced at the bearing pass frequency. The figure below shows the measured spectrum for GBX_A17_02 on 12 March. The dominant peak is the gear mesh frequency at 312 Hz, and it is flanked by symmetric sidebands spaced at 14.2 Hz, the calculated bearing pass frequency for this unit; the first, second and third order sidebands are all present, with amplitude falling as the order rises. That pattern is diagnostic. Broadband energy alone was still modest when those sidebands were already unambiguous, which is exactly why NW_STD_0417 defines the alarm on the sideband criterion in addition to an overall level, and why the overall level alone gave so little notice.

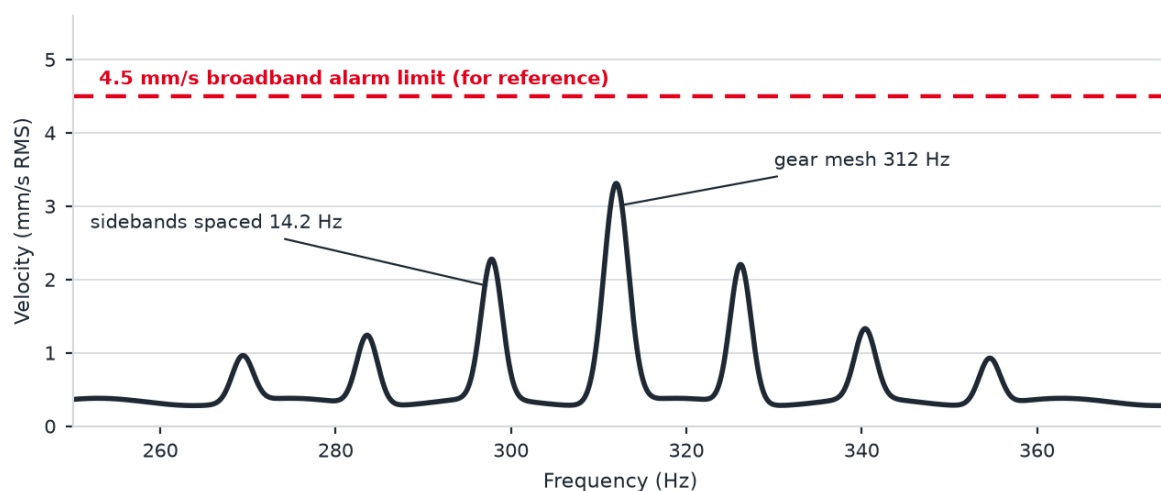


Figure 1. Velocity spectrum at the high speed stage of GBX_A17_02, 12 March 2026. The dashed line is the NW_STD_0417 BROADBAND alarm limit of 4.5 mm/s RMS, shown for reference only: it applies to the quadrature sum across the band, not to any single spectral line. Note that no line approaches it while the sideband family is already unambiguous.

Contributing factor: the casting batch

The failure occurred well short of design life, so the analysis looked for a contributing factor rather than treating it as a random event. Material certificates trace the planet carrier and raceway of GBX_A17_02 to GBX_BATCH_2023_11, casting run 11 of 2023. Hardness testing of the retained sample for that batch returned values at the low end of the specified range, and coupons taken from the failed raceway itself confirmed the same. Raceway hardness is not a cosmetic property here. It sets the depth and magnitude of the subsurface shear stress the material tolerates before micro cracks initiate, so a lower hardness shortens the initiation phase described above. It does not change the mechanism and it does not change the spectral signature; it changes the clock. The conclusion is that GBX_A17_02 failed by an ordinary and well understood mechanism on an accelerated schedule, and the accelerant is a material property shared by every unit from the same casting run. That is the finding with fleet consequences, because GBX_BATCH_2023_11 was not a single unit purchase. This report deliberately does not enumerate the other affected units: the asset register is the authoritative record of batch membership and it should be queried rather than transcribed into a report that will age.

Corrective and preventive actions

Immediate: GBX_A17_02 is removed and replaced, and the replacement unit is drawn from a different casting run. The oil system was flushed and the filters replaced, because circulating debris from a spalled raceway is an initiation source for the replacement unit if it is left in place. Short term: every gearbox traceable to

GBX_BATCH_2023_11 is placed on a shortened inspection interval with spectral review rather than overall level review, and the sideband criterion in NW_STD_0417 is applied explicitly at each review. Turbine WTG_A17 was not the only unit from that run and the register should be consulted to establish the current list, including units whose overall readings are still nominal, since this report has established that a nominal overall level is not evidence of a healthy raceway. Longer term: incoming hardness verification is added as a receiving inspection step for gearbox castings rather than relying on the supplier certificate alone, and the condition monitoring configuration is updated so that the appearance of pass frequency sidebands raises a notification in its own right, independently of the broadband trend.